Hello

Threaded Binary Tree (TBT)

What is Threaded Binary Tree?

- A Binary Tree in which **NULL pointers are replaced by threads**.
- A thread points to the **inorder predecessor** or **inorder successor** of a node.
- Each node has two extra tags to indicate whether the left/right link is a real child or a thread.

Node Structure



ltag = 0 → left is a child
ltag = 1 → left is a thread

rtag = 0 → right is a child
rtag = 1 → right is a thread

Why is it Used?

Avoids Null Pointers
Utilizes the otherwise unused NULL links.

Traversal Without Stack/Recursion
Inorder, preorder or postorder traversal can be done without using extra space.

Space Efficient
No auxiliary stack or recursion call stack required.

Faster Traversal
Reduces overhead of stack operations.

Where is it Used?

Expression Trees
In compilers to handle expression parsing and evaluation.

Symbol Tables
Efficient inorder traversal is useful in compiler design.

Memory Efficient Traversal
Useful in systems with memory constraints (embedded systems).

General Tree Operations
Used wherever frequent inorder traversal is required without extra space.

code

Insertion
&
Traversal

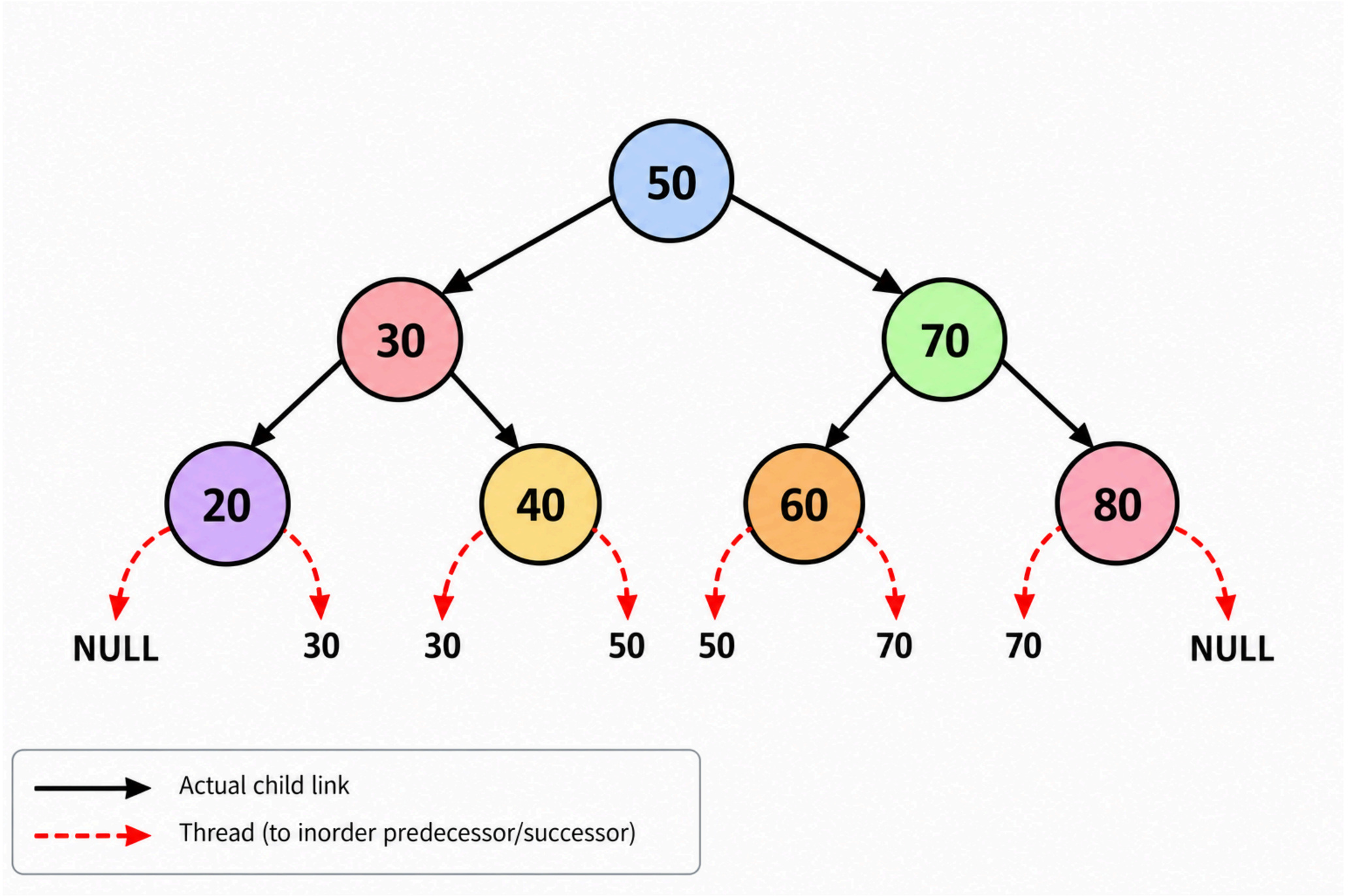
Step 1 — Header Files

Step 2 — Define the Node Structure

Step 3 — Create the TBT Class

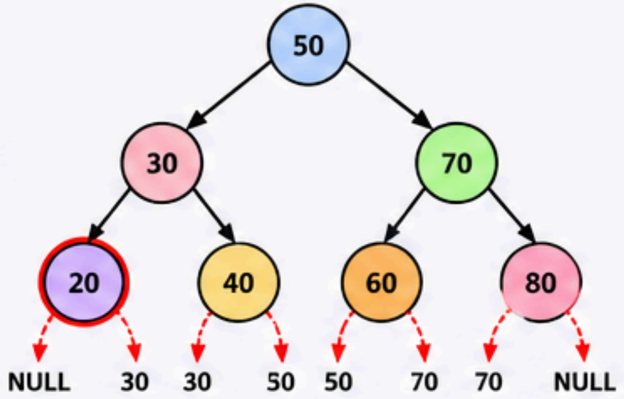
Step 4 — Initialize the Tree

Step 5 — Implement Insertion


50
30
70
20
40
60
80
NULL
30
30
50
50
70
70
NULL
Actual child link
Thread (to inorder predecessor/successor)

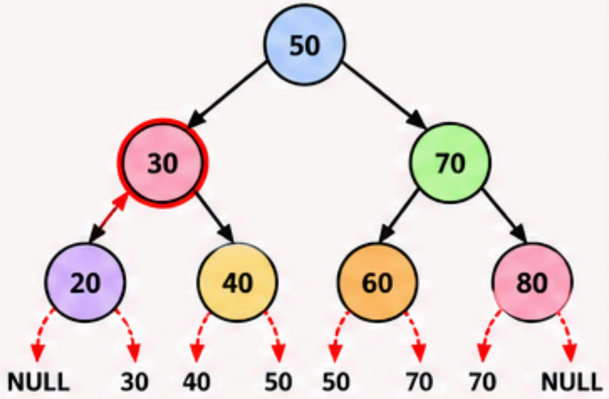
Step 6 — Implement Inorder Traversal

1 Start at leftmost node



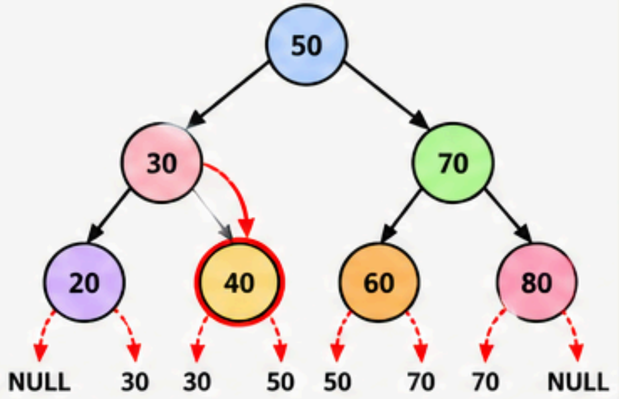
Go to leftmost node of the tree.
Current node = 20
Print 20

2 Move using right thread



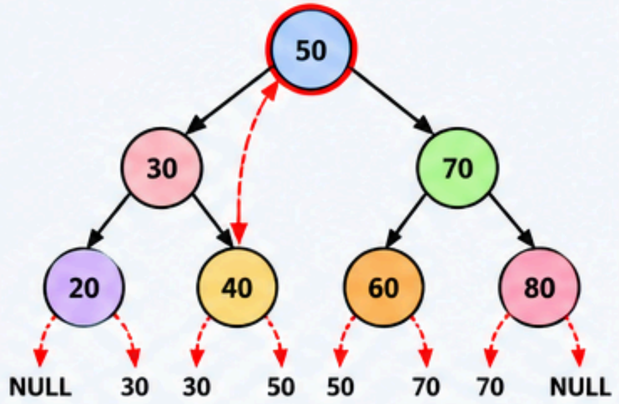
20 has rthread = true
Move to its inorder successor (right thread).
Current node = **30**
Print 30

3 Move to leftmost of right subtree



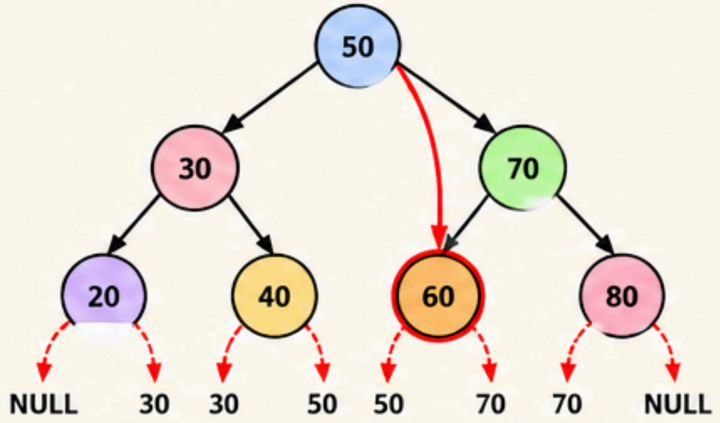
30 has rthread = false
Go to leftmost node of right subtree.
Current node = 40
Print 40

4 Move using right thread



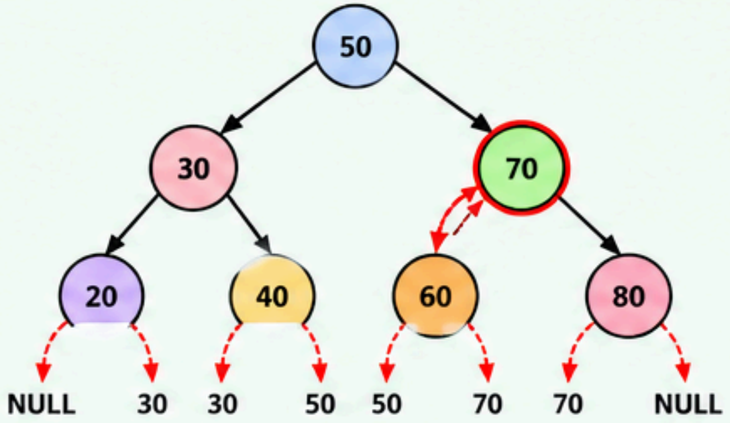
40 has rthread = true
Move to its inorder successor (right thread).
Current node = **50**
Print 50

5 Move to leftmost of right subtree



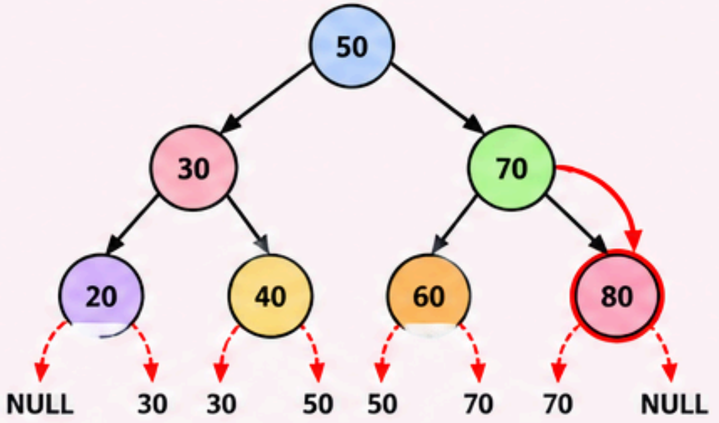
50 has rthread = false
Go to leftmost node of right subtree.
Current node = 60

6 Move using right thread



60 has rthread = true
Move to its inorder successor (right thread).
Current node = 70

7 Move to leftmost of right subtree



70 has rthread = false
Go to leftmost node of right subtree.
Current node = 80
Print 80

Step 7 — Write main()

Thank you